

ZIYAO CHEN

EDUCATION

Northwestern Polytechnical University

M.Eng. in Computer Technology

Admitted with exemption from the National Entrance Examination

Key Laboratory of Big Data Storage and Management, Ministry of Industry and Information Technology

GPA: 85.93/100

Xi'an, China

Sep. 2024 - Jun. 2027

Advisor: Prof. Lingyun Song

First-Class Scholarship; Outstanding Graduate Student

Hefei University of Technology

B.Eng. in Computer Science and Technology

School of Computer and Information

GPA: 85.76/100

Hefei, China

Sep. 2020 - Jun. 2024

First-Class Scholarship; Outstanding Student

RESEARCH INTERESTS

Continual Learning, Self-Evolving Agents, Multimodal Large Language Models, Knowledge Editing

PUBLICATIONS

- Lingyun Song, **Ziyao Chen**, Kang Pan, et al. KSS-MoE: Knowledge Space Synergy Framework in Mixture of Experts for Continual Visual Instruction Tuning. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- Ziyao Chen**, Xuequn Shang, and Lingyun Song. Iteratively-Enhanced Dynamic Expansion Model for Continual Learning. *Web Information Systems and Applications (WISA) 2024*. (CCF-C; Oral Paper).

PATENTS

- Lingyun Song, **Ziyao Chen**, Xuequn Shang. An Iteratively-Enhanced Dynamic Expansion Continual Learning Model. CN120123765A, Feb. 2025 (Published).
- Ziyao Chen**, Hai Min, Yemao Zhang, Kai Shu, Han Chen. A Facial Landmark Detection Method Based on Mixed-Domain Attention Mechanism. CN115223228B, Aug. 2022 (Granted)
- Xuequn Shang, Binze Shi, Lingyun Song, **Ziyao Chen**, et al. Information Diffusion Prediction Method for Hybrid Graph Transformer Based on 2D Positional Encoding Enhancement. CN120541410A, May 2025 (Published)

RESEARCH EXPERIENCE

Research on Multi-Step Reasoning Methods for Complicated Visual Problems under Data Constraints

Graduate Researcher

NSFC General Program; Jan. 2026 - Dec. 2029

- Led the continual adaptation direction for data-constrained multi-step visual reasoning, studying how multimodal models can acquire new knowledge without eroding prior reasoning capabilities.
- Designed risk-aware orthogonal knowledge-subspace updates that isolate new edits from previously learned subspaces, protecting stability-critical knowledge during long update sequences.
- Developed multi-granularity knowledge integration and semantic activation mechanisms to retrieve edited knowledge selectively while preserving model behavior on unrelated inputs.

Step-Level DPO for MoE LLMs in Continual Learning of Complex Multi-Domain Tasks

Lead Graduate Researcher

Xiaomi Open-Competition Research Program; May 2025 - Jul. 2026

- Developed KSS-MoE for continual visual instruction tuning, using internal activations to fuse task-relevant low-rank subspaces from specialized experts for fine-grained collaboration at inference.
- Designed specialized and shared expert collaboration with conflict masks and orthogonal gradient projection to facilitate cross-task transfer while reducing negative interference.
- Evaluated the framework across all eight CoIN tasks, achieving state-of-the-art continual-learning performance; the work was published at **AAAI 2026**.

Multimodal Intelligent Continual Question Answering with Dynamic-KG RAG

Graduate Researcher

CCF-Zhipu Large Model Innovation Fund; Nov. 2024 - Nov. 2025

- Investigated coordinated adaptation between evolving structured graph knowledge and multimodal model knowledge for temporal and cross-domain learning.
- Constructed visual, textual, multimodal, and shared expert subspaces to strengthen cross-domain knowledge transfer while retaining previously acquired knowledge.
- Introduced cross-subspace orthogonality constraints to decouple domain-specific and general knowledge optimization, improving generalization while mitigating forgetting and conflicts.

HONORS & AWARDS

- National First Prize, National Computer Competition for Colleges and Universities Dec. 2022
- Provincial First Prize, Chinese Collegiate Computing Competition (4C) (*Project Lead*) Aug. 2022
- Provincial First Prize, 16th iCAN Innovation Contest (*Project Lead*) Sep. 2022
- Provincial Silver Award, China International College Students' Innovation Competition (*Project Lead*) Jul. 2025
- Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) Mar. 2023
- Provincial Undergraduate Innovation Program (*Project Lead*) May 2021 - Apr. 2022
- Recipient of **more than ten** national, provincial, and ministerial-level awards, including eight as project lead.